

Cloud data storage and management tools

In this case study you're a data analyst at a gaming company tasked with evaluating user engagement with in-game advertisements by gathering data from multiple sources to help the advertising team decide whether to continue the current ad-supported model or transition to a paid subscription strategy.

Let's look at each of the different data sources.

Cloud Storage

The first data management tool you use is Cloud Storage. This will **host all data** once it's compiled. With this tool, you can **upload data from remote servers** through an internet connection. While working toward your business question, you're able to move your application databases, analytics, and data science tools into one place.

Dataflow

You also use Dataflow in this project to **create a data pipeline**. Dataflow gives you the option of either **stream or batch processing**. This makes it possible to **stream** data from your various apps into one database. You can also use the batch processing feature to import any data you already stored on your computer.

Cloud Data Fusion

Cloud Data Fusion is the tool you use to build a data pipeline. You can also **create** data integrations, **run** high-volume data pipelines, and **continuously** integrate user data from your applications into your database.

BigQuery

Once you have data from all sources streaming into Cloud Storage, you use BigQuery to **write Sequel queries to join data** and **clean the data** to ensure it's complete, has no duplicates, and will be useful for your project.

Managed Service for Apache Spark

Next, Managed Service for Apache Spark allows you to **use open-source data analytics tools**

at a large scale. And you can **apply programming languages and algorithms to data** also at a large scale. With these tools, you're able to move your data to the next phase and start drawing conclusions.

After analyzing the data using the relevant tools, you share your discoveries with the advertising team, highlighting how many users click on ads and what percentage of total gamers that equals. With this information, the advertising team can effectively choose whether to keep using ads to generate revenue or charge users a fee.

You've just experienced a typical business case involving data management and storage. Along the way, you successfully integrated multiple tools to complete a data workflow and find important answers for your stakeholders.

Cloud data professionals are excellent data stewards—and, with the strategies from this example case, you now understand how to use key tools and solutions to PROTECT and CONTROL all kinds of data!